PADHAKU
ICT – Class 6 · Class 6 ICT



CHAPTER 2

Data Representation and Processing–01 (Spreadsheet)

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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⭐ Introduction

? **Answer these questions about yourself:**

What is your name? What is your age? How many members are in your family?

Draw your picture. Which is your favourite song? Which is your favourite movie?

💡 SAMPLE ANSWER

Write your own answers. Example:

- Name: **Riya**
- Age: **11 years**
- Family members: **5**
- Favourite song: **Vande Mataram**
- Favourite movie: **Taare Zameen Par**

All these answers are **data** about you.

🧪 Activity 1

1 **Mother sent Rohan to the market. The shopkeeper gave him a bill. Look at the bill and find out what data it contains.**

💡 SAMPLE ANSWER

The bill contains this data (check the picture in your book):

- Shop name and address – **text data**
- Customer name (Rohan) – **text data**
- Bill number – **numeric data**
- Date of the bill – **date data**

- Names of items – **text data**
- Quantity, rate, amount and total – **numeric data**

⭐ Introduction to Data Types

? Sahil, can you give an example of text data?

✔ ANSWER

Text data is made of letters and numbers of any language.

Examples: **Name: Ramesh**, **Address: A 20 Mohan Nagar**.

? A date is also made of numbers. Is it numeric data?

✔ ANSWER

No. A date is a different type of data – **date data**.

It is written in a fixed format, Day-Month-Year, e.g. 10 November 2022 is written **10-11-2022**, not 22-11-10.

? What type of data are photos and videos?

✔ ANSWER

A photo is **image data**.

A video is **video data**.

? **Are the songs we hear on FM also data?**

✔ ANSWER

Yes. Songs and recordings of voice are called **audio data**.

🧪 Activity 2

2 **The picture shares data about some children. Identify the different data types in it.**

💡 SAMPLE ANSWER

(check the picture in your book)

- Name, school, city – **text data**
- Age, class, roll number, marks – **numeric data**
- Date of birth – **date data**
- Photo of the child – **image data**
- A voice recording or song – **audio data**
- A video clip – **video data**

🧪 Activity 3

3 **Find things around you that look like a spreadsheet and write their names.**

💡 SAMPLE ANSWER

Things made of rows and columns:

- School attendance register
- Class time table

- Mark sheet / report card
- Railway or bus time table
- Calendar
- Electricity bill or shop bill

⭐ Practice - 1

1 **In the given spreadsheet:**
(a) Write the address of the marked cells inside the cells.
(b) Colour (shade) the cells G12, H5, D19 red.

💡 SAMPLE ANSWER

(a) A cell address = **column letter + row number**. Find the column letter at the top and the row number at the left of each marked cell, e.g. a cell in column C and row 4 is **C4** (check the picture in your book).

(b) Click cell **G12** → click the **Background Color** button on the Formatting toolbar → choose **Red**.

Do the same for **H5** and **D19**.

🧪 Activity 4

4 **Store a list of difficult words and their meanings from the first chapter of your English book in a spreadsheet.**

💡 SAMPLE ANSWER

Open a blank worksheet. In row 1 type the headings: **S.No | Word | Meaning**. Type the words from row 2 onwards and save the file. Example:

S.No	Word	Meaning
1	glisten	shine
2	gentle	soft, kind
3	curious	eager to know
4	whisper	speak very softly

🧪 Activity 5

5 **Make your class time table in a spreadsheet.**

💡 **SAMPLE ANSWER**

Write the periods in row 1 and the days in column A. Example:

Day	Period 1	Period 2	Period 3	Period 4
Monday	English	Maths	Science	Hindi
Tuesday	Maths	Hindi	Social Science	ICT
Wednesday	Science	English	Maths	Sanskrit

Save the worksheet with a name like **TimeTable**.

Activity 6

6 **In the planets worksheet you made, insert a new column before the Moons column and write the colour of every planet in it.**

💡 **SAMPLE ANSWER**

- Click any cell of the **Moons** column (column C).
- Click **Insert → Columns**. A new column comes on its left.
- Type the heading **Colour** and the colours:

Planet Name	Colour
Mercury	Grey
Venus	Yellowish white
Earth	Blue
Mars	Red
Jupiter	Orange-brown
Saturn	Pale yellow
Uranus	Light blue
Neptune	Dark blue

Activity 7

7 **In the planets worksheet, insert a new row at the top and write in it: Our Planets.**

✔ **ANSWER**

- Click any cell of **row 1**.

- Click **Insert → Rows**. A new row comes above it.
- Click cell **A1** and type **Our Planets**. Press Enter and save.

Activity 8

8 **Write the names of five plants found near your home and their height. Find the highest, the lowest and the average height, and the sum of all heights.**

💡 **SAMPLE ANSWER**

Example (heights in cm) in cells A1:B6:

Plant	Height (cm)
Tulsi	60
Rose	90
Money plant	150
Aloe vera	40
Hibiscus	120

- Highest: **=max(B2:B6) → 150**
- Lowest: **=min(B2:B6) → 40**
- Average: **=average(B2:B6) → 92**
- Sum: **=sum(B2:B6) → 460**

Activity 9

9 **Look at the map of India and make a table:**

- Choose any five rivers and write their names.
- Write the state where each river starts.
- Write their lengths.
- Find the longest length, the shortest length and the sum of all lengths.

💡 SAMPLE ANSWER

River	Starts in (State)	Length (km)
Ganga	Uttarakhand	2525
Yamuna	Uttarakhand	1376
Godavari	Maharashtra	1465
Narmada	Madhya Pradesh	1312
Krishna	Maharashtra	1400

If the lengths are in C2:C6:

- Longest: **=max(C2:C6)** → **2525 km** (Ganga)
- Shortest: **=min(C2:C6)** → **1312 km** (Narmada)
- Sum: **=sum(C2:C6)** → **8078 km**

❓ Exercise · Multiple Choice Questions

1 **The data made of letters and numbers of any language is called:**

a. Text b. Numeric c. Audio d. Video

✔ ANSWER

a. Text – text data is made of letters and numbers of any language.

2 **Identify the correct cell address:**
a. A7 b. 77 c. 7A d. 7_A

✔ ANSWER
a. A7 – column letter first, then row number.

3 **A function always starts with the sign:**
a. % b. * c. & d. =

✔ ANSWER
d. = – every function starts with the = sign, e.g. =sum(B2:B6).

4 **How is data organised in a spreadsheet?**
a. Paragraph b. Rows and Columns c. Images d. Video

✔ ANSWER
b. Rows and Columns – a spreadsheet is a grid of rows and columns.

5 **A recording of a song or voice is called _____ data.**
a. Text b. Numeric c. Audio d. Video

✔ ANSWER

c. Audio

6 **The _____ function is used to add the data in a cell range.**

a. Count b. Sum c. Max d. Min

✔ ANSWER

b. Sum – =sum() adds the values in the range.

❓ Exercise · Short Answer Questions

1 **What is a spreadsheet? Write the names of any two popular spreadsheet software.**

✔ ANSWER

A spreadsheet is a computer application that helps to **store a large amount of data** in rows and columns and to **do calculations** on it.

Examples: **Microsoft Excel**, **Google Sheets** (also OpenOffice Spreadsheet, LibreOffice Calc).

2 **What is a worksheet? Write its use.**

✔ ANSWER

A worksheet is a **single sheet** of rows and columns in a spreadsheet. Many worksheets together make a **workbook**.

Use: we store text, numbers and dates in its cells in an orderly way and find results like total, highest, lowest and average with functions.

3 **Define a cell.**

✔ ANSWER

The meeting point (intersection) of a **row** and a **column** is called a **cell**. Data is stored in a cell.

Every cell has an address, e.g. **A1** (column A, row 1). The cell with a thick dark border is the **active cell**.

4 **Write the benefits of storing data in a spreadsheet.**

✔ ANSWER

- Large amounts of data can be stored neatly in rows and columns.
- Calculations like sum, average, max, min and count are quick and correct.
- Data can be changed, added (rows/columns) or deleted easily.
- The file can be saved and used again; no long paper lists.

❓ Exercise · Long Answer Questions

1 **Ankita made a list of the classes in her school (Column A) and the number of children in each (strength, Column B) in a spreadsheet. Write the steps to:**

a. Show the highest strength in cell B10

b. Show the lowest strength in cell B11

c. Show the sum of all strengths in cell B12

d. Show the average strength in cell B13

e. Show the total number of classes in cell B14

💡 SAMPLE ANSWER

Suppose the strengths are in cells **B2 to B9** (check the picture in your book), so the cell range is **B2:B9**.

a. Click cell **B10**, type **=max(B2:B9)** and press Enter.

b. Click cell **B11**, type **=min(B2:B9)** and press Enter.

c. Click cell **B12**, type **=sum(B2:B9)** and press Enter.

d. Click cell **B13**, type **=average(B2:B9)** and press Enter.

e. Click cell **B14**, type **=count(B2:B9)** and press Enter.

(You can also write labels like "Maximum Strength" in A10 to A14.)

2 **Hari made a list of the temperature of every month of his state (months in Column A, temperature in Column B) in a spreadsheet. Write the steps to show:**

a. the highest temperature in cell B10

b. the lowest temperature in cell B16

c. the sum of all temperatures in cell B17

d. the average temperature in cell B18

e. the total number of months in cell B19

💡 **SAMPLE ANSWER**

(The book says "strength" and "classes" in b–e; here it means temperature and months.)

Suppose the 12 months are in rows 2 to 13, so the range is **B2:B13** (check the picture in your book). If a result cell already has data, use any empty cell instead.

a. Click cell **B10**, type **=max(B2:B13)** and press Enter.

b. Click cell **B16**, type **=min(B2:B13)** and press Enter.

c. Click cell **B17**, type **=sum(B2:B13)** and press Enter.

d. Click cell **B18**, type **=average(B2:B13)** and press Enter.

e. Click cell **B19**, type **=count(B2:B13)** and press Enter → **12**.

🎉 **Chapter 2 done! Revise the answers once more tomorrow.**